MCA SOFTWARE ENGINEERING LAB

Experiment No. 6: Sequence Diagrams for Shield AI

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Project Title	Shield AI — Scam Detection System	Date	28 Oct 2026

Aim

To prepare sequence diagrams for important functions of the student's own software project and understand the order of interaction between users and system objects.

Task

Choose THREE important scenarios from your project. Example: Login, Add Record, Search, Payment, Book Issue, Return Book, Generate Report, etc. Prepare one sequence diagram for each scenario.

Part A - Identify the Scenarios

No.	Scenario / Function	Main Actor	Purpose
1	Scan a message	End user	Submit text and receive a scam-risk verdict with explanation.
2	Check URL reputation	Analysis engine	Retrieve URL/domain reputation where an external service is configured.
3	Submit feedback	End user	Record optional feedback for a previous scan.

Part B - Identify Participants for Each Scenario

Write the actor(s) and objects/lifelines that take part in each scenario.

Scenario	Actor(s)	Objects / Lifelines	Database involved?
Scan a message	End user	User interface, Analysis Engine, Result Module, History Store	Optional scan history
Check URL reputation	Analysis Engine	URL Adapter, Reputation API	No local DB required; external service
Submit feedback	End user	User interface, Feedback Module, History Store	Yes, if feedback storage is enabled

Part C - Write Messages in Correct Order

Before drawing, write the messages from top to bottom. Use meaningful names such as validateUser(), searchItem(), saveRecord(), showResult().

Scenario 1: Scan a message

Step	Sender -> Receiver	Message
1	User → UI	submitText(text)
2	UI → Analysis Engine	analyse(text)
3	Analysis Engine → Result Module	return riskSignals and score
4	Result Module → History DB	saveScan(metadata), if history is enabled
5	Result Module → UI	show verdict and explanation

Scenario 2: Check URL reputation

Step	Sender -> Receiver	Message
1	Analysis Engine → URL Adapter	checkURL(url)
2	URL Adapter → Reputation API	lookupDomain(url)
3	Reputation API → URL Adapter	return reputation response / status
4	URL Adapter → Analysis Engine	return normalised URL risk result
5	Analysis Engine → Result Module	combine URL signal with text signals

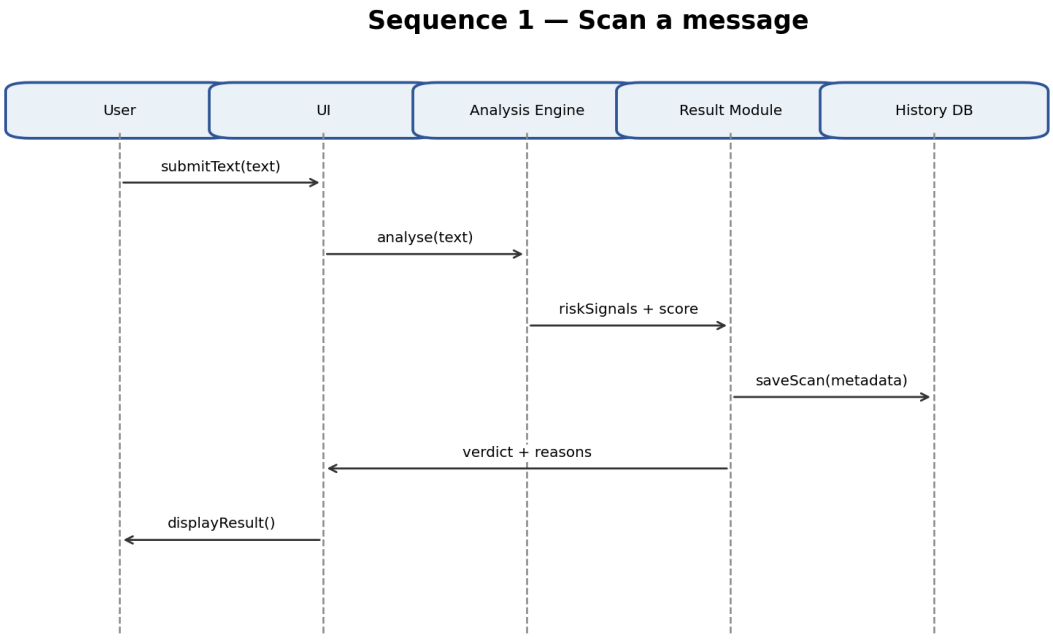
Scenario 3: Submit feedback

Step	Sender -> Receiver	Message
1	User → UI	submitFeedback(scanId, label)
2	UI → Feedback Module	validateFeedback(data)
3	Feedback Module → History DB	storeFeedback(scanId, label)
4	History DB → Feedback Module	return saved confirmation
5	Feedback Module → UI	showThankYou()

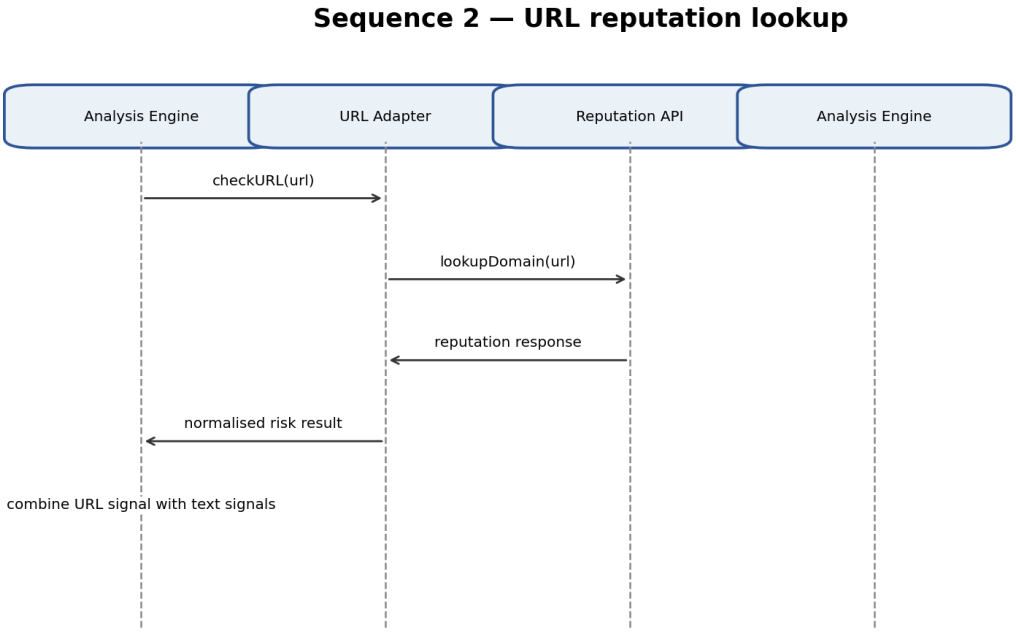
Part D - Draw the Sequence Diagrams

Draw THREE sequence diagrams. Show actor/object names, lifelines, messages in time order, and use alt or loop only when needed.

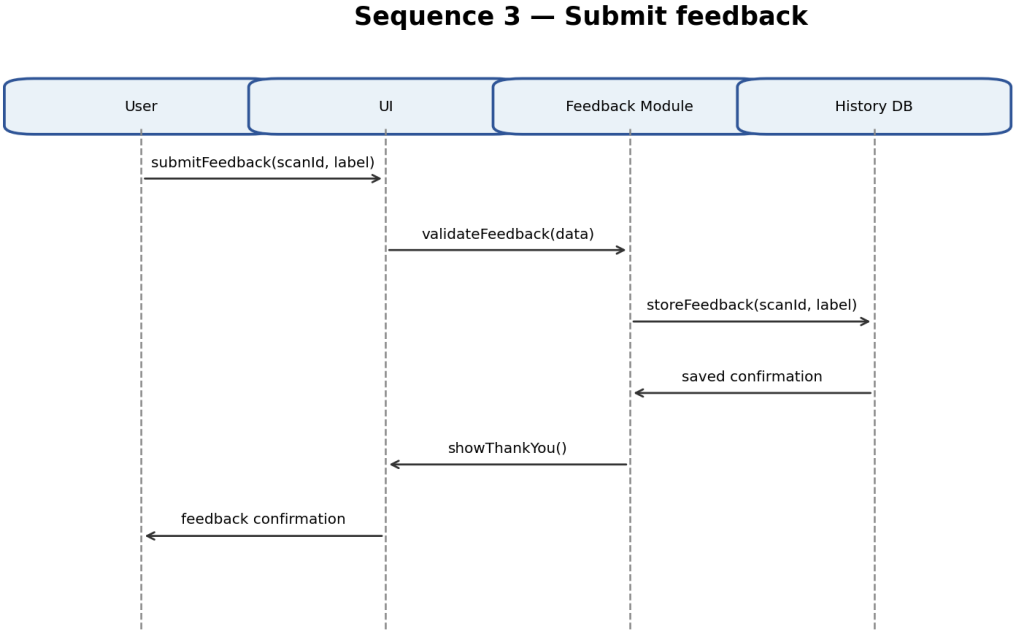
Sequence Diagram 1: Scan a message



Sequence Diagram 2: Check URL reputation



Sequence Diagram 3: Submit feedback



Part E - Check Your Work

☐ Messages are in correct time order.

☐ Actor and all necessary lifelines are shown.

☐ Message names are clear and meaningful.

☐ alt is used for alternatives/conditions where required.

☐ loop is used for repeated actions where required.

☐ The diagram matches the requirements of our project.

Result

In this experiment, we prepared sequence diagrams for scanning a message, checking URL reputation and submitting feedback in Shield AI. We identified the actors, system objects and messages and represented their interaction in the correct order.